



$$2k \times \binom{t}{2} x_i x_j y_i$$

$$k t (t-1) x_i x_j y_i$$

$$k(k-t)(t-1) y_i - \boxed{y_k}$$

$$- (k-t)(t-1) + \binom{t-1}{2}$$

$$k \rightarrow k(k-t)(t-1) H_k + k t (t-1) H_1$$

$$k-1 \rightarrow k(k-t)(t-1) H_{k-1} + (2k-1)(t-1) H_2 + \left((t-1) + (2k-1) \binom{t-1}{2} \right) \cdot H_1 + O \times \binom{t-1}{2}$$

$$(k-t)(t-1) + \binom{t-1}{2} > \frac{(2k-1)(t-1)}{2}$$

$$\binom{t-1}{2} > (t-1) \left(\left(k - \frac{1}{2} \right) - (k-t) \right)$$

$$\frac{t-2}{2} < \left(t - \frac{1}{2} \right)$$

$$\frac{t}{k} \mid 2k \binom{t}{2}$$

$$k t \mid k(k-t)(t-1)$$

$$\binom{t-1}{2} \quad 1 \quad 0$$

$$(t-1) \quad 2k-1 \quad 1$$

Handwritten scribbles and marks.

$x_i x_j$

$$18H_6 + 18$$

$$18H_5 + 10H_2 + 7$$

$$18H_4 + 10H_2 + 4H_3 + 4$$

$$\begin{matrix} 2 & \frac{1}{3} & \frac{4}{6} & \frac{2}{3} \\ & & & \\ & & & \\ & & & \\ & & & \\ & & & \\ & & & \\ & & & \\ & & & \\ & & & \end{matrix}$$

$$\frac{18}{5} + \frac{13}{5}$$

$$1 + \frac{4 \times 5}{6} \left(\frac{26}{6} \right)$$

u_t $32t$

$$x_i \uparrow \quad y_i \downarrow 2tx\left(\frac{t}{2}\right) \quad x_i x_j \quad y_i \leftarrow b$$

$$\downarrow \frac{1}{k} \quad 2tx\left(\frac{t}{2}\right) y_1 - y_{2t} \leftarrow a$$

$$n \quad 2t^2(t-1) H_a + \left(\frac{b}{2}\right) a H_3 + \left(b(t-b)\left(2t-a\right) + \left(\frac{t-b}{2}\right) a\right) H_1 + b(t-b)a H_2$$

$$\frac{\left(\frac{t-b}{2}\right) \times 2t}{4tx\left(\frac{t}{2}\right)} \geq \frac{t-b}{2t} \Leftrightarrow \frac{(t-b)(t-b-1)}{t(t-1)} \geq \frac{t-b}{t}$$

$$\Leftrightarrow b=0$$



EIR
✓

core
X

1 C_1
3 d_i C_1 C_2

$V_{1/2}$
 $\frac{1}{3} \frac{1}{2}$
 $\frac{1}{3} \frac{1}{2}$
 $\frac{1}{3} \frac{1}{2}$

→ 2 $d_1 d_2 d_3$ $2 \frac{10}{6} + 3 \frac{1}{2}$

$\rho_{\text{eff}} = |S| = 4$ d_i $C_1 = 4$
 $|T| = 1$ 2

$k \leftrightarrow m$

$|T| >$

$\frac{k}{2} + 1$

$\frac{k}{2} + 1$

k $\frac{k}{2}$ x

$\frac{k}{2}$ y

$\frac{k}{2}$

$\binom{k}{\frac{k-1}{2}}$

$x_1 - x_2$

$x_1 x_2 x_3$

$x_1 - 1$

15

15

15

94

$15H6 + 15H2 + 3H3$

$15H6 + 15H3 + 3H5$

$15H3 + 15H2 + 3H5$

$15H2 + 15H3 + 3H5$

$15H3 + 15H4 + 3H5$

$15H4 + 15H5 + 3H6$

$15H5 + 15H6 + 3H7$

$15H6 + 15H7 + 3H8$

$15H7 + 15H8 + 3H9$

$15H8 + 15H9 + 3H10$

$15H9 + 15H10 + 3H11$

$15H10 + 15H11 + 3H12$

Approval CC

JR ✓

$$|S| \geq \frac{n}{k}$$

$$c_i \in \cap A_i$$

$$q \rightarrow \left(\frac{n}{k}\right) \text{ انفرادی}$$

$$\frac{n - \frac{n}{k}}{k} \rightarrow \text{مجموعی}$$

PJR ✗

$$l=2 \quad k=3$$

$$\{c_1, c_2\}$$

$$c_3$$

$$c_4$$

$$q = |S| \geq \frac{2}{3} \cdot 6 = 4$$

$$|A_i| \geq 2$$

$$U A_i = \{c_1, c_2\}$$

$$JR < PJR \text{ مثال برای } \square$$

Preserve EJR

PJR ✓

EJR ✗

$$c_1, c_2, c_3$$

$$d_4$$

$$d_3$$

$$d_2$$

$$d_1$$

$$d_0$$

$$d_{-1}$$

$$d_{-2}$$

$$d_{-3}$$

$$d_{-4}$$

$$d_{-5}$$

$$d_{-6}$$

$$d_{-7}$$

$$d_{-8}$$

$$d_{-9}$$

EJR 15

PAV

$$|S| \geq \frac{n}{k}$$

$$|A_i| \geq l$$

$$|A_i \cap W| < l$$

$$|A_i \cap W| < l$$

$$|A_i \cap W| < l$$

$$|A_i \cap W| < l$$

$$|A_i \cap W| < l$$

$$|A_i \cap W| < l$$

$$|A_i \cap W| < l$$

$$|A_i \cap W| < l$$

$$|A_i \cap W| < l$$

$$|A_i \cap W| < l$$

$$\frac{n|T|}{k^2} \leq |S| \quad S'$$

$$8 \quad |A; \cap T| > |A; \cap W|$$

$$|T| \rightarrow \frac{1}{|T|} \times |S| \geq \frac{n}{k}$$

$$c_1 \dots c_k \quad \boxed{k - |T|}$$

$$\frac{n - |S|}{k}$$

SD

$$k=5$$

$$|T|=3$$

~~SD~~

$$\left(\frac{n - |S|}{k} \right) |T| \leq \frac{1}{|T|} \times |S|$$

$$\frac{n}{k} \leq \frac{1}{|T|} \times |S|$$

$$\left(\frac{n}{k} - \frac{n}{k^2} |T| \right) |T| \leq \frac{n}{k}$$

$$\frac{2}{5} \left(1 - \frac{|T|}{k} \right) |T| \leq 1$$

$$\frac{|T|}{k} + 1 - \frac{|T|}{k} = 1$$

$$\frac{k}{2} \left(1 - \frac{1}{k} \right)$$

$$\frac{k}{2}$$

5	a	c ₁	c ₂	c ₃
2	b	c ₁		
2		c ₂		
2		c ₃		
2		c ₄		
2		c ₅		

$$\frac{a}{a+5b} = \frac{3}{5}$$

$$2a = 15b$$

$$a = 15$$

(2)

$C_1 - C_k$

$C_1 - C_k$

$d_1 - d_{k+1}$
 $d_1 - d_{k+1}$
 $d_1 - d_{k+1}$

0

$\frac{1}{2}$

$\frac{1}{2}$



$$\frac{k}{2} \binom{k}{2} \binom{k+1}{2}$$

$$= bk$$

$H_{k+1} - H_k$

$$\frac{t}{k}$$

$$\frac{k + (t-1)}{2}$$

$$k \binom{t}{2} x_i x_j y_i$$

$$\frac{k(k-t)(t-1)}{2} y_i - y_k$$

$$k(k-t)(t-1)$$

$k=4$
 $a \quad b \quad c \quad d \rightarrow e \quad f$
 $a \quad b \quad e \quad f$

$a \quad e \quad f$
 $b \quad e \quad f$
 $c \quad e \quad f$
 $d \quad e \quad f$
 $e \quad f$
 PAV
 $\frac{k}{2}$

ab
 $\frac{1}{6}$

$|T|=2$

$\rightarrow \frac{4}{2}$

$$\frac{6}{12} + \frac{6}{11} + \frac{5}{11}$$

$\frac{6}{5} \rightarrow b$
 $\frac{12}{6} \rightarrow 20$
 $c_3 \rightarrow c_{12} \rightarrow \frac{1}{20}$

$$c_1 \rightarrow \frac{1}{8}$$

$$c_2 \rightarrow \frac{1}{13}$$

$$a, b \rightarrow \frac{1}{4}$$

$$\frac{1}{8} + \frac{1}{13} + \frac{1}{2}$$

$$\frac{1}{4} + \frac{1}{13}$$

$$\frac{a+c}{a+b}$$

$$\frac{b-c}{a+b}$$

$$\frac{(b-c)}{(a+b)b} > \frac{(a+c)}{(a+b)c(a+1)}$$

$$b > \frac{a+1}{a} \frac{c^2}{c}$$

$$c=2 \quad 4$$



$$k W_{k-a} > \frac{(a+c)}{c} \left(\frac{W_{a+1} + C W_{k-a}}{c} \right)$$

$$\frac{a}{c} X +$$

$$\frac{a}{c} X + c Y$$



PAV

$$\frac{1}{1} > \frac{1}{2} > \frac{1}{3} \dots$$

$$W_1 > W_2 > \dots$$

$$\frac{b-c}{a+b} W_b > \frac{a+c}{(a+b)c} W_{a+1}$$

$$b W_b > c W_b + \frac{a+c}{c} W_{a+1}$$

$$(b-c) c W_b > (a+c) W_{a+1}$$

$$(a-c+1) c > (a+c)$$

$$c=2$$

$$a=99$$

$$b=100$$

$$200 > 101$$

$$(k-(a+c)) c W_{k-a} > (a+c) W_{a+1}$$

$$k=9 \quad a=4 \rightarrow 9-(4+4) = 1$$

$$(5-c) c > 4+c$$

$$k c W_{k-a} > (a+c) (W_{a+1} + c W_{k-a})$$

$$\frac{(a+b)c}{b} > (a+c) \left(\frac{b}{a+1} + \frac{c}{b} \right)$$

$$b((a+1)-a+c) > (a+c)(a+1)c - a(a+1)$$

$$b(1-c) > (a+1)(c^2 + ac - a)$$

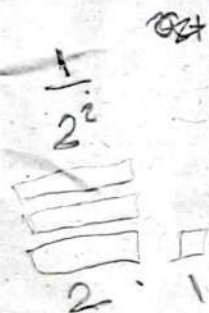
وچو

PAV

:cwo1-13

$$k W_{k-a} > (a+c) \left(\frac{W_{a+1}}{c} + W_{k-a} \right) \rightarrow k, \text{ مبرور، قابل}$$

$$k > (a+c) \left(\frac{W_{a+1}}{(W_{k-a})c} + 1 \right) \quad a+c \leq k$$



$$2^{k-1}$$

$$k \leq (a+c) \left(\frac{\epsilon^{1-k}}{c} + 1 \right)$$

$$k >$$

$$k \leq (a+1) \left(\frac{W_{a+1}}{W_{k-a}} + 1 \right) \quad \frac{c+1}{c} \epsilon^{1-k} + c$$



$$\frac{k-1}{k} W_k \leq W_1$$

$$k \leq (a+c) \left(\frac{W_{a+1}}{W_{k-a}c} + 1 \right)$$

$$(k-1) W_k \leq W_1$$

$$k \leq (k-1-a-c) \left(\frac{W_{k-a}}{W_{a+1}c} + 1 \right) \rightarrow ?$$

$$W_k < \frac{1}{k-1}$$

$$k^2 \leq$$

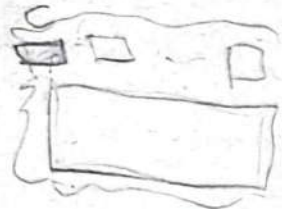
$$\frac{k-a}{k}$$

$$\epsilon^k$$

$$\epsilon^k = \frac{1}{k}$$

$$b > \frac{a+1}{a} \quad \frac{c^2}{c-1}$$

$$c=4 \quad a=2 \quad b=17 \quad 998$$



$$\frac{a+c}{a+b}$$

$$\frac{b-c}{a+b}$$



$$\frac{1}{6} \quad \frac{6}{19} \quad 0.006$$

$$\frac{13}{19}$$

$$0.994 \quad \frac{1}{k}$$

$$\frac{998}{994}$$

$$\frac{2}{6}$$

$$994 + 6$$

$$994 \left(\frac{996}{994} \right)^2 + \frac{8}{3}$$

$$994 \left(\frac{2}{994} + \frac{998}{994} \right)$$

max min
min max

$$\frac{b-c}{a+b}$$

b-

a bc



15

k

994 (

$$994 \left(\frac{995}{994} \right)^2 + 3x$$

CE T

C' E W/T

$\geq \frac{n}{k}$

$$\sum x_i^2 \quad \frac{k}{n}$$

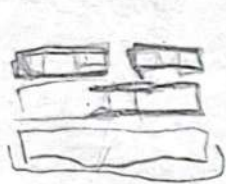
(T)

C'

IDK

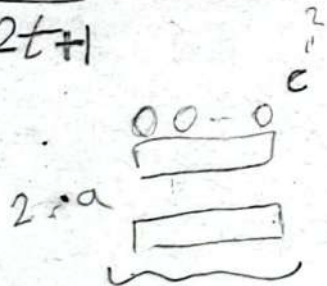
$$k=10 \quad 3 \quad \frac{7}{10} \quad \frac{4}{10}$$

$$n=20 \quad \sim \sim \sim \sim$$



$$c/a \quad 2/3 \quad 5/7$$

$$\frac{t+s}{2t+1}$$



$$\frac{a+c}{b+c}$$

$$(a+c)c$$

$$(b-a)c$$

$$\frac{b-a}{b}$$

$$\frac{12}{27} = \frac{4}{9}$$

$$16 \quad 13$$

$$\frac{4}{9} \quad \frac{10}{27}$$

$$(b-a)(a+1) < b(a+c)$$

$$b(ac-a) > ac(a+1)$$

$$ba(c-1) > ac(a+1)$$

$$\frac{b(c-1)}{c} > (a+1)$$

$$b > \frac{c}{c-1} (a+1)$$

$$c-2 > b > \frac{10}{9} (a+1)$$

$$b > \frac{10}{9} (a+1)$$

$$k=a+b$$

$$\frac{b-a}{b} > \frac{a+c}{c^2}$$

$$\frac{(b-a)c}{b} > \frac{a+c}{a+1}$$

$$c^2(b-a) > b(a+c)$$

$$a+b+c > b(c-a)$$

$$b(c^2-c-a) > 4a$$

$$b(2-a) > 4a$$

$$a=1 \quad b=5$$

$$\frac{2+2}{5}$$

$$|A_i \cap W| < |A_i \cap T| \rightarrow \text{PAV}$$

$$C \in T/W \quad W \cup \{c\}$$

$$\sum x_i^2$$

c'

$$8 \times 160$$

$$\frac{1}{12} \times 96 = 8$$

$$\frac{b-c}{b(a+b)} < \frac{1}{a+b}$$

$$\frac{a+c}{a+b}$$

$$\frac{b-c}{a+b}$$

$$c(a+c) \quad \frac{a+c}{a+b}$$

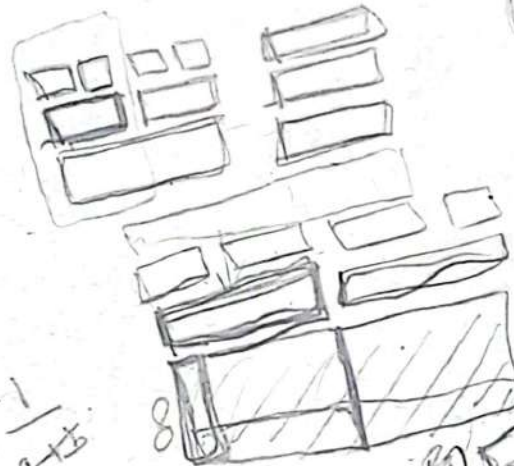
$$c(b) \quad \frac{b-c}{a+b}$$

$$c=2$$

$$\frac{b-c}{a+b} \quad \frac{b-1}{a+b}$$

$$2\sqrt{A}$$

$$n \frac{H}{k} \frac{r}{r_1}$$



$$26 \frac{8}{16}$$

$$\frac{3}{8} \quad \frac{8}{16} \quad \frac{105}{16}$$

$$\frac{1}{k}$$



$$\frac{m}{64}$$

$$\frac{1}{16}$$

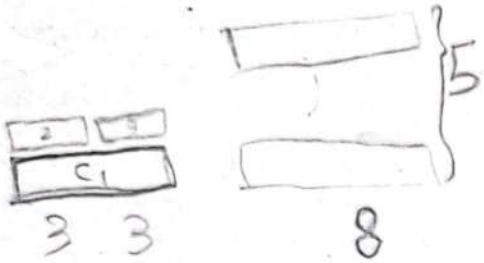
$$b=8$$

$$a=8$$

$$c=1$$

$$k=16$$

PAV



$$\frac{a+c}{a+b}$$

$$\frac{8}{5} \quad \frac{3}{2} \quad \frac{3}{6}$$

$$b > \frac{a+1}{a} \frac{c^2}{c-1}$$

$$c=2$$

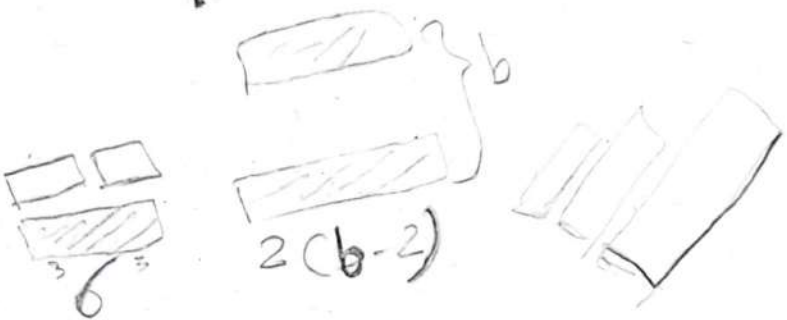
$$a=1$$

$$b>8$$

$$4$$

$$c(a+c) \quad c(b-c)$$

$$\frac{c(b-c)}{b} > \frac{a+c}{a+1}$$



$$2 \frac{(b-2)}{b} > \frac{3}{2}$$

$$4b$$

$$a=8$$

$$b=8$$



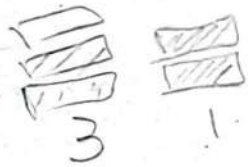
$$20 \quad \frac{1}{20}$$

$$b(ac+c-a-c) > c^2(a+1)$$

$$ba(c-1) > c^2(a+1) \quad c=2$$

$$c=1 \quad a=2 \quad b=2$$

$$\frac{1}{2} \quad \frac{3}{2}$$



$$x_1 \ x_2 \ x_3 \ \binom{6}{2}$$

$$y_1 \sim y_6$$

$$15 H_6 + 15 H_2 \rightarrow \frac{15}{6}$$

$$15 H_5 + 10 H_3 + 5 H_2 \rightarrow \frac{10}{3}$$

$$\rightarrow 15 H_4 + 6 H_4 + 8 H_3 + H_2$$

$$x_i$$

$$x_i x_j \ y_i \rightarrow 6$$

$$x_i x_j x_k \ y_i y_j$$

x

$$y_1 \sim y_6 \rightarrow 6$$

$$3 H_6 \quad 3 + 3 H_3$$

$$+ H_{2t}$$

$$+ H_{2t-1} + 1$$

$$t H_{2t-2} + 2$$

$$t H_t + t$$

$$k \cdot 2t$$

$$\binom{2t}{t-1} x_1, \dots, x_t \ y_i - y_{i,t-1}$$

$$\binom{2t}{t-1} \rightarrow y_1 \dots y_{2t}$$

$$\binom{2t}{t-1} (H_{2t} + H_{t-1}) > \binom{2t}{t-1} H_t + \sum \binom{t}{i} \binom{t}{i+1} H_{t-i}$$

$$\binom{2t}{t-1} H_{2t-1} + \binom{2t-1}{t-1} H_t + \binom{2t-1}{t-2} H_{t-1}$$

$$\frac{\binom{2t}{t-1}}{2t} > \frac{\binom{2t-1}{t-1}}{t}$$

CET

$$\frac{n}{k} |T| \leq |S|$$

Remember

$$\frac{b-c}{(a+b)b} > \frac{1}{(a+b)} \frac{1}{a+1}$$

$$> \frac{a+1}{a} \frac{c^2}{c-1}$$

$$\sum x_i^2$$

$$CET/W \rightarrow$$

$$\frac{n}{k}$$

$$\frac{k}{n}$$

$$\frac{k}{n}$$

$$\frac{a+c}{a+b}$$

$$\frac{a}{k}$$

$$a=1$$

Core X

$$\frac{a}{k}$$

$$C' \in W$$

$$\uparrow$$

$$T$$

$$n$$

$$\frac{n}{k}$$

$$\frac{n}{k}$$

$$(T)$$

$$\epsilon$$

$$\begin{aligned} \epsilon a_i &= A \\ \epsilon c_i &= C \\ A+B+C &= m \\ A+B &= k \end{aligned}$$

$$\frac{a+c}{a+b}$$

$$\frac{a_2+c_2}{A+B}$$

$$\frac{a+c}{A+B}$$

$$EJR$$

$$\frac{B-C}{A+B}$$

$$\frac{m}{2}$$

$$k=3$$

Don't forget to check the table!

$$\begin{aligned} &PAV \\ &Phygen \\ &I \end{aligned}$$

$$\sum x_i^2$$

$$\frac{k}{n}$$

$$\frac{1}{a+b}$$

$$\frac{1}{a+b}$$

$$PAV$$

$$A$$

$$\sum x_i^2$$

$$\frac{\sum x_i^2}{k^2}$$

$$\frac{1}{n}$$

$$\frac{1}{n}$$

$$\frac{1}{n}$$

we have so many conditions

1 ✓
2 →

5 2^C $\binom{m}{C}$
C $m \times A_i$

$T = \{c_1\}$

$|T| = 1 \rightarrow EJR$

$|T| = \{c_1, c_2\}$

$|S| \geq \frac{2}{k}$

PAV

$|A_i| \ll k$

$a=2$ ①

① $\frac{b}{a+b}$

$\frac{1}{b}$

$\frac{b-c}{b}$

$\frac{a-c}{(a+b)a+b}$

c_1, c_2, c_3

T_1, T_2

PAV → phragmen

⇒

مجموعه

افزاد

T_1, T_2

EJR

T_1, T_2

T

minimal

کوچکترین مجموعه که شامل همه باشد

w

k

گروه با هم که کمتر از k باشد